



NOUHA ESSID

Software Engineer (Deep Learning Focus)

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Portfolio: nouhaessid.github.io/my-portfolio

SUMMARY

Computer Science Engineer specialized in Deep Learning and Computer Vision, with hands-on experience developing machine learning models using Python, TensorFlow, and PyTorch. Strong understanding of CNNs and image processing, supported by academic and personal projects. Familiar with NLP techniques for text analysis.

WORK EXPERIENCE

Mitacs Globalink Research Internship **Ontario Tech University** *Oshawa, Canada* **06/2024 - 12/2024**

- Applied Deep Reinforcement Learning algorithms (DQN, TD3) using TensorFlow and OpenAI Gym to optimize navigation strategies and compared them against traditional algorithms (A*, Artificial Potential Field).

Apprenticeship Machine Learning **NextAV** *Sousse, Tunisia* **01/2023 - 08/2023**

- Implemented a **Random Forest classifier** in **Python** to predict crop types using moisture and vegetation indexes, achieving improved model accuracy through **k-fold cross-validation** and feature optimization

EDUCATION

Engineering Degree in Computer Science *Sousse, Tunisia* **09/2021 - 12/2024**
National Engineering School of Sousse (ENISo)

Pre-Engineering Program – Math & Physics (MP Track) *Monastir, Tunisia* **09/2019 - 06/2021**
Monastir Preparatory Engineering Institute (IPEIM)

PROJECTS

- Facial Verification System:** Implemented a CNN-based Siamese network in **Python** (**TensorFlow**, **Keras**) to compare facial embeddings and measure similarity scores for identity verification.
- Handwritten Digit Recognition:** Developed a digit classification system using the **MNIST dataset**, implementing two models — a fully connected neural network and a **convolutional neural network (CNN)** — both built with **Python**, **Keras**, and **TensorFlow**.
- House Price Prediction:** Developed a **regression model** with **Scikit-learn** to predict housing prices applying **data preprocessing**, **feature scaling** and **model evaluation** techniques.
- Sentiment Analysis (Zero-Shot Model):** Applied **Hugging Face** zero-shot classification pipeline for sentiment analysis. Implemented simple inference pipeline demonstrating use of **pretrained NLP models**.

SKILLS

- Programming Languages: Python, Java, JavaScript
- Libraries & Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, Optuna, OpenAI Gym, TensorBoard, BeautifulSoup
- Machine Learning Techniques: Deep Learning, Random Forest, SVM, Transfer Learning, Computer Vision, CNN, NLP
- Tools & Platforms: Git, Jupyter, Matplotlib, Docker
- Other: Data Preprocessing, Model Evaluation, Problem solving, Web Scraping

LANGUAGES

- English – Fluent
- French – Fluent
- Arabic – Native

CERTIFICATIONS

- **Deep Learning Specialization** – DeepLearning.AI (Coursera)
Courses: Neural Networks and Deep Learning, Improving Deep Neural Networks, Structuring Machine Learning Projects, Convolutional Neural Networks
- **TOEIC Certificate** – ETS Global
- Angular from Zero to Hero – Instructor BOUALI ALI
- **Docker Training Course** – KodeKloud